EXP 10

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1. Aim

The aim of this project is to design and develop an Inventory Management Dashboard using HTML, CSS, and Chart.js that provides a visual representation of product stock levels and inventory category distribution.

The dashboard enables users to:

- Monitor the current stock quantities of various products (Laptops, Mobiles, Headphones, Keyboards, and Monitors)
- Understand the distribution of inventory across different categories (Electronics, Accessories, and Peripherals)
- Make data-driven decisions through interactive bar and pie charts
- View a clean and responsive interface suitable for business use

2. Procedure

The following steps were followed to build the Inventory Management Dashboard:

1. Set up the HTML file (visual.html) with the basic document structure including <head> and <body> tags.
2. Included the Chart.js library via CDN (https://cdn.jsdelivr.net/npm/chart.js) inside the <head> section for chart rendering.
3. Applied CSS styling to the body, container, and canvas elements to ensure proper layout, background color (#f4f6f8), centered alignment, and responsive canvas sizes (max-height: 400px, max-width: 600px).
4. Created a containing two <canvas> elements: one with id='barChart' for the bar chart, and one with id='pieChart' for the pie chart.
5. Defined sample inventory data in JavaScript:

- Products array: ['Laptops', 'Mobiles', 'Headphones', 'Keyboards', 'Monitors']
- Quantities array: [50, 120, 75, 40, 30]
- Categories array: ['Electronics', 'Accessories', 'Peripherals']

- Category counts array: [200, 100, 50]

6. Implemented a Bar Chart using Chart.js by calling new Chart() on the barChart canvas context, setting type to 'bar', supplying labels as products, and quantities as dataset data. Different background colors were assigned to each bar.
7. Implemented a Pie Chart using Chart.js on the pieChart canvas context with type set to 'pie', using categories as labels and categoryCount as data, with three distinct background colors (#36A2EB, #FF6384, #FFCE56).
8. Configured chart options for both charts including responsive: true, hidden legend for the bar chart, and visible title text ('Product Stock Levels' and 'Inventory Category Distribution').
9. Saved the file as visual.html and opened it in the browser using PowerShell command: powershell -Command "Start-Process 'c:/Users/DELL/Desktop/visual.html'"
10. Verified the output in the browser to confirm both charts rendered correctly with proper labels, colors, and data.

3.Screenshots

The following screenshots illustrate the development environment and the final rendered output of the Inventory Management Dashboard:

Screenshot 1: Dashboard Output - Bar Chart & Pie Chart

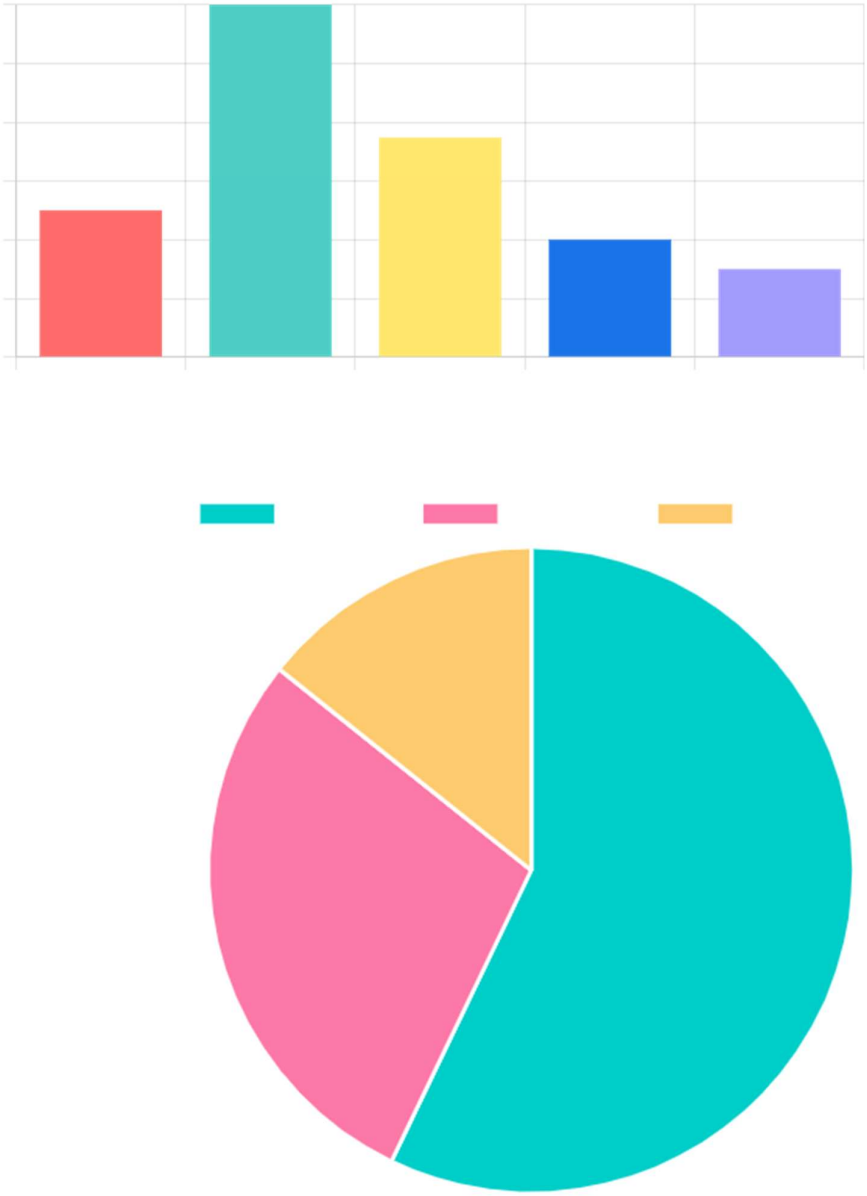

Figure 1: Rendered Inventory Management Dashboard showing Product Stock Levels (Bar Chart) and Inventory Category Distribution (Pie Chart)

Screenshot 2: HTML Code - Structure, Styles, and Bar Chart Configuration

visual.html

C: > Users > DELL > Desktop > visual.html > html > body > script

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Inventory Dashboard</title>
    <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
    <style>
        body {
            font-family: Arial;
            background: #f4f6f8;
            text-align: center;
        }
        .container {
            width: 60%;
            max-width: 800px;
            margin: auto;
        }
        canvas {
            max-height: 400px;
            max-width: 600px;
            margin: 20px auto;
            display: block;
        }
    </style>
</head>
<body>

    <h1>📦 Inventory Management Dashboard</h1>

    <div class="container">
        <canvas id="barChart"></canvas>
        <canvas id="pieChart"></canvas>
    </div>

    <script>
        // Sample Inventory Data
        const products = ["Laptops", "Mobiles", "Headphones", "Keyboards", "Monitors"];
        const quantities = [50, 120, 75, 40, 30];

        const categories = ["Electronics", "Accessories", "Peripherals"];
        const categoryCount = [200, 100, 50];

        // 📊 Bar Chart (Product Quantities)
        const barCtx = document.getElementById('barChart').getContext('2d');
        new Chart(barCtx, {
            type: 'bar',
            data: {
                labels: products,
                datasets: [{
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS E:\vs code\html> powershell -Command "Start-Process 'c:/Users/DELL/Desktop/visual.html'"
PS E:\vs code\html>
```

Figure 2: VS Code showing the HTML structure, CSS styling, Chart.js CDN import, and Bar Chart initialization code

Screenshot 3: JavaScript Code - Bar Chart & Pie Chart Implementation

visual.html

C: > Users > DELL > Desktop > visual.html > html > body > script

```
2   <html lang="en">
26  <body>
35      <script>
43          // 📊 Bar Chart (Product Quantities)
44          const barCtx = document.getElementById('barChart').getContext('2d');
45          new Chart(barCtx, {
46              type: 'bar',
47              data: {
48                  labels: products,
49                  datasets: [{
50                      label: 'Stock Quantity',
51                      data: quantities,
52                      backgroundColor: [
53                          '#4CAF50',
54                          '#2196F3',
55                          '#FFC107',
56                          '#FF5722',
57                          '#9C27B0'
58                      ]
59                  }]
60              },
61              options: {
62                  responsive: true,
63                  plugins: {
64                      legend: { display: false },
65                      title: {
66                          display: true,
67                          text: 'Product Stock Levels'
68                      }
69                  }
70              }
71          });
72
73          // 🥧 Pie Chart (Category Distribution)
74          const pieCtx = document.getElementById('pieChart').getContext('2d');
75          new Chart(pieCtx, {
76              type: 'pie',
77              data: {
78                  labels: categories,
79                  datasets: [{
80                      data: categoryCount,
81                      backgroundColor: [
82                          '#36A2EB',
83                          '#FF6384',
84                          '#FFCE56'
85                      ]
86                  }]
87              },
88              options: {
89                  responsive: true,
90                  plugins: {
91                      title: {
92                          display: true,
93                          text: 'Inventory Category Distribution'
94                      }
95                  }
96              }
97          });
98      </script>
99
100 </body>
101 </html>
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS E:\vs code\html> powershell -Command "Start-Process 'c:/Users/DELL/Desktop/visual.html'"
PS E:\vs code\html>
```

Figure 3: VS Code showing complete Chart.js configuration for both the Bar Chart (Product Quantities) and Pie Chart (Category Distribution)